

MSPM0_K210 Object Detection

1. K210 and MSPM0 Communication

1.1 Experimental Prerequisites

This tutorial uses the MSPM0G3519 development board. K210 needs to run the program in **K210-AI(MSPM0)** to start the experiment.

MSPM0G3519 *1

K210 vision module *1 (with SD card (with AI model) and camera)

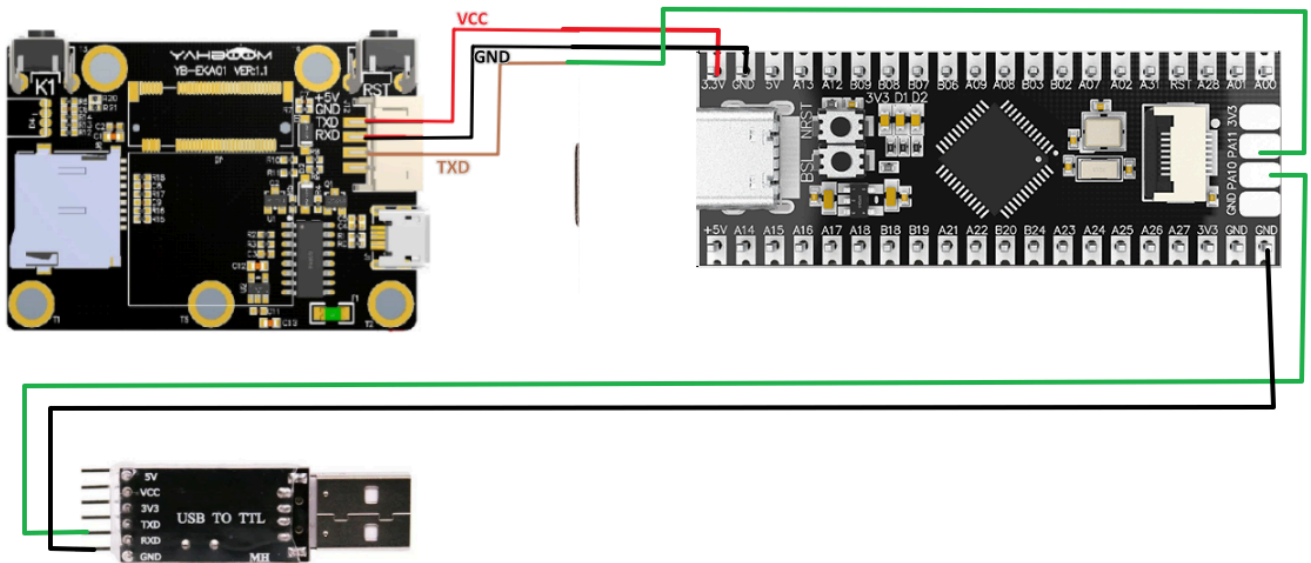
USB to TTL module *1

1.2 Experimental Wiring

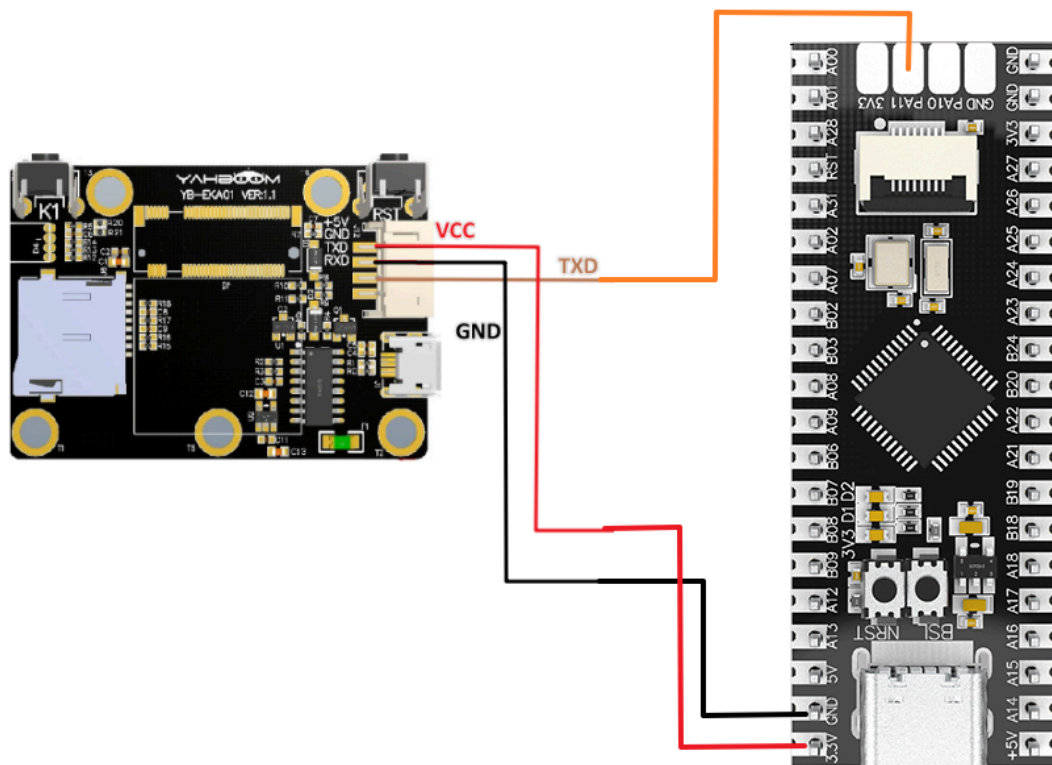
Method 1: Using USB to TTL module

MSPM0G	K210 Vision Module
RX	TXD
GND	GND
VCC	5V

MSPM0G	USB to TTL Module
TX	RXD
GND	GND



Method 2: Using onboard Type-C port



1.3 Main Code Explanation

This example sets the serial port print baud rate to 115200 bps, and the K210 module connection baud rate is set to **115200** bps.

Type Filter Text... X <<

PROJECT CONFIGURATIO...
Project Config... 1/1 ✓ +

MSPM0 DRIVER LIBRARY ...
SYSTEM (9)
Board 1/1 ✓ +
DMA +
GPIO +
MATHACL +
Configuration NVM +
RTC +
SYSCTL 1/1 ✓ +
SYSTICK 1/1 ✓ +
WWDT +
ANALOG (6)
ADC12 +
COMP +
DAC12 +
GPAMP +
OPA +
VREF +
COMMUNICATIONS (6)
I2C +
I2C - SMBUS +
MCAN +
SPI +
UART 2/4 ✓ +
UART - LIN +
TIMERS (6)
TIMER - CAPTURE +

Software > UART

UART (2 of 4 Added) ⓘ
+ ADD REMOVE ALL
✓ MYUART
✓ K210_Uart
Name MYUART
Selected Peripheral UART0
Quick Profiles
UART Profiles Custom
Basic Configuration
UART Initialization Configuration
Clock Source BUSCLK
Clock Divider Divide by 1
Calculated Clock Source 32.00 MHz
Target Baud Rate 115200
Calculated Baud Rate 115211.52
Calculated Error (%) 0.01
Word Length 8 bits
Parity None
Stop Bits One
HW Flow Control Disable HW flow control

Type Filter Text...
X
<<
<
>
Software > UART

PROJECT CONFIGURATIO...

Project Config... 1/1 ✓ (+)

MSPM0 DRIVER LIBRARY ...

SYSTEM (9)

Board 1/1 ✓ (+)

DMA (+)

GPIO (+)

MATHACL (+)

Configuration NVM (+)

RTC (+)

SYSCTL 1/1 ✓ (+)

SYSTICK 1/1 ✓ (+)

WWDT (+)

ANALOG (6)

ADC12 (+)

COMP (+)

DAC12 (+)

GPAMP (+)

OPA (+)

VREF (+)

COMMUNICATIONS (6)

I2C (+)

I2C - SMBUS (+)

MCAN (+)

SPI (+)

UART 2/4 ✓ (+)

UART - LIN (+)

TIMERS (6)

TIMER - CAPTURE (+)

TIMER - COMPARE (+)

UART (2 of 4 Added) ②

✓ MYUART

✓ K210_Uart

Name

K210_Uart

Selected Peripheral

UART1

Quick Profiles

UART Profiles

Custom

Basic Configuration

UART Initialization Configuration

Clock Source

BUSCLK

Clock Divider

Divide by 1

Calculated Clock Source

32.00 MHz

Target Baud Rate

115200

Calculated Baud Rate

115211.52

Calculated Error (%)

0.01

Word Length

8 bits

Parity

None

Stop Bits

One

HW Flow Control

Disable HW flow control

```

int main(void)
{
    SYSCFG_DL_init();

    NVIC_ClearPendingIRQ(MYUART_INST_INT_IRQN); //Clear serial port interrupt flag
    NVIC_EnableIRQ(MYUART_INST_INT_IRQN); //Enable serial port interrupt

    while (1)
    {
        if (k210_msg.class_n != 0) //Example number is not empty
        {
            if(k210_msg.class_n == 9) //Object detection
            {
                sprintf(buff_com, "x=%d,y=%d,w=%d,h=%d\r\n", k210_msg.x, k210_msg.y, k210_msg.w, k210_msg.h);
                uart0_send_string(buff_com);
            }
        }
    }
}

```

```

        sprintf(buff_com,"id = %s\r\n",k210_msg.msg_msg);
        uart0_send_string(buff_com);

        k210_msg.class_n = 0;//Clear example number
    }

}
delay_ms(500);

}
}

```

After the above program runs this example, the k210_msg structure members will have corresponding values and will be printed via serial port.

k210_msg: is the structure for receiving information. Its main members are:

- x: is the x-coordinate of the upper-left corner of the recognized box (range: 0-240)
- y: is the y-coordinate of the upper-left corner of the recognized box (range: 0-320)
- w: is the width of the recognized box (range: 0-240)
- h: is the height of the recognized box (range: 0-320)
- id: is the recognized label
- class_n: example number
- msg_msg[20]: valid data

After data reception and processing, each member of k210_msg will store valid information. If you want to do secondary development, you can directly call the members of k210_msg.

Note: The Keil project source code must be placed under the SDK path for compilation.

1.4 Experimental Phenomenon

1. After connecting the cables, the K210 vision module runs offline [K210 offline running method](#)
2. Set the serial assistant interface as shown in the figure

Port
COM4: USB-SERIAL CH34C

Baud rate 115200

Stop bits 1

Data bits 8

Parity None

Operation  Close

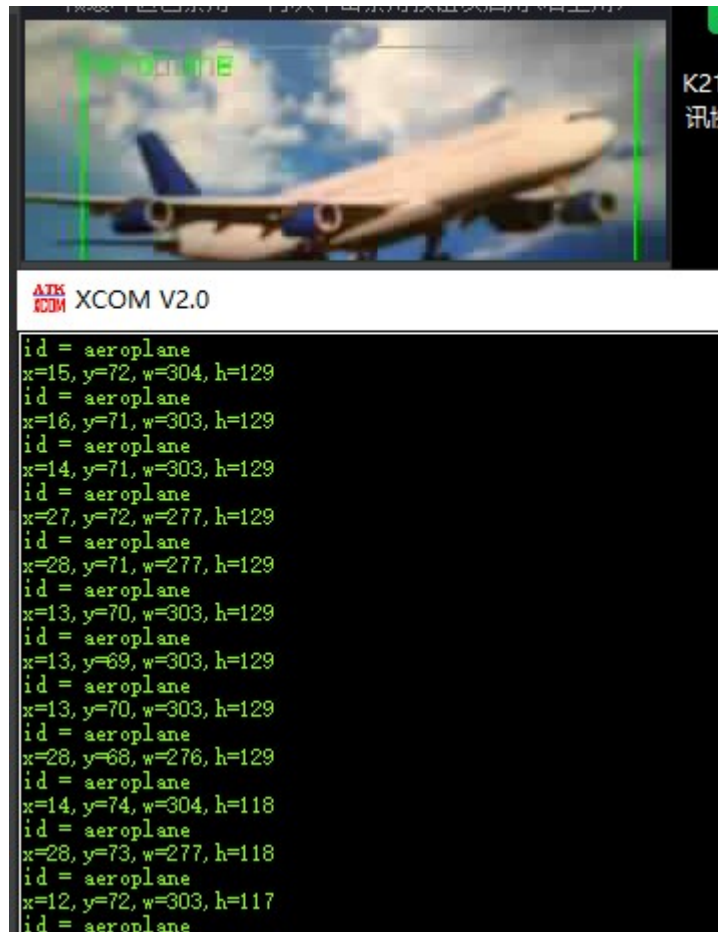
Save Data Clear Data

☐ Hex ☐ DTR

☐ RTS ☐ 自动保存

☐ TimeStamp 1000 ms

- Then run the object detection example. The serial assistant will print the important information transmitted by K210 to MSPM0G, as shown in the figure below.



Object detection only transmits the 6 member variables of k210_msg: x, y, w, h, id, msg.

id: The information content is (0.airplane, 1.bicycle, 2.bird, 3.boat, 4.bottle, 5.bus, 6.car, 7.cat, 8.chair, 9.cow, 10.dining table, 11.dog, 12.horse, 13.motorcycle, 14.person, 15.potted plant, 16.sheep, 17.sofa, 18.train, 19.monitor)